

Hailey A. Reed

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Education

University of Connecticut

Storrs, CT

Bachelor of Science, Major in Computer Science, Minor in Mathematics • GPA: 3.6/4.0

2022-2026

- Concentration in Computational Data Analytics • Honors Program • UConn STEM Scholar Community.
- Relevant Coursework: Graduate-Level Machine Learning, Artificial Intelligence, and Principles of Databases.

Research Experience

Undergraduate Research Assistant

Storrs, CT

Project: *DeltaSegment: Multifidelity Medical Image Segmentation via Change Detection*

Jan 2025 – Present

Advisor: Dr. Qian Yang | Scientific & Computational Machine Learning Laboratory (SciMaLL)

- Co-developed DeltaSegment, a multi-fidelity, change-detection-based Semi-Siamese U-Net framework that improves small-data medical image segmentation by leveraging high-low fidelity image pairs generated via color quantization.
- Co-authored a paper accepted to IEEE ISBI 2026, contributing to model design, experiments, and quantitative analysis.

Undergraduate Research Assistant

Storrs, CT

Project: *GDP Nowcasting using FRED-MD Macroeconomic Indicators*

May 2025 – Present

Advisor: Professor Joseph Johnson

- Reproduced the GDP nowcasting framework from Németh & Hadházi (2023), implementing benchmark neural architectures (MLP, 1D-CNN, Elman RNN, LSTM, GRU) on the FRED-MD macroeconomic panel.
- Built a regime-switching ensemble that identifies macroeconomic conditions and selects the optimal model, with full walk-forward validation, feature engineering, and a Plotly Dash dashboard for interpretability.

Publications

Hailey A. Reed, Yushuo Niu, Qian Yang. “DeltaSegment: Multifidelity Medical Image Segmentation via Change Detection,” Submitted to 2026 IEEE 23rd International Symposium on Biomedical Imaging (ISBI), Paper ID: 1571226357.

Leadership & Innovation

Entrepreneurial Lead – Accelerate UConn (NSF I-Corps) Cohort 32

Hartford, CT

UConn Center for Entrepreneurship & Innovation

June 2025 – July 2025

- Selected as Entrepreneurial Lead for UConn’s NSF I-Corps cohort, translating a lab-developed AI change-detection model into an industrial defect-inspection product called LiteVision through customer discovery, stakeholder outreach, and competitive landscape analysis.
- Conducted TAM/SAM/SOM market sizing, business-model development, and product-market validation to evaluate commercialization and feasibility and identify early adopter segments for AI-driven manufacturing inspection tools.

Projects

AlarmCast – Senior Design Project (Team Lead)

Sept 2025 – Present

- Leading a 5-member team developing a low-cost Raspberry Pi smoke/CO alarm listener and full backend-mobile pipeline for real-time alerts and verification, using a Kanban workflow to coordinate hardware, backend, and mobile integration.

Personal Portfolio Website – <https://haileyannreed.github.io/portfolio/>

Mar 2025

- Curated and deployed a personal site showcasing projects in AI, ML, computer vision, and quantitative finance.

Technical Skills

ML & CV: PyTorch, TensorFlow, Scikit-Learn, NumPy, Pandas • Transformers, RNN/LSTM/GRU • U-Net, Siamese/Semi-Siamese Models, Data Augmentation

Programming & Tools: Python, SQL, C, Haskell, Git, LaTeX, SQLite, REST APIs, Security (Argon2), HPC (UConn)

Work Experience

BEACH (Belonging, Engagement, Affinity Computer Hangout) Lifeguard

Aug 2024 - Present

UConn School of Computing, Storrs, CT

- Tutoring undergraduate students in computer science fundamentals; Organizing faculty-student engagement events.